

資料庫文檔

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概述

ClassroomIO 使用 **PostgreSQL 15.x** 作為主資料庫，通過 **Supabase** 平台託管。資料庫包含 37 個表格、2 個視圖、4 個自定義枚舉類型和 15+ 個存儲函數，支援多租戶數據隔離。

技術特性

特性說明

- PostgreSQL 版本15.x
- 表格數量37
- 視圖數量2 (lesson_versions, dash_org_stats)
- 枚舉類型4 (LOCALE, PLAN, COURSE_TYPE, COURSE_VERSION)
- 存儲函數15+
- 擴展套件uuid-openssl, pgcrypto, pgjwt, moddatetime, vector
- 安全機制Row Level Security (RLS) + 5 個 RLS 輔助函數
- 即時訂閱Supabase Realtime
- 自動 APIPostgREST

RLS 輔助函數

函數說明

`is_org_admin(org_id)` 檢查用戶是否為組織管理員

`is_org_member()` 檢查用戶是否為組織成員

`is_user_in_course_group(group_id)` 檢查用戶是否在課程群組中

`is_user_in_group_with_role(group_id)` 檢查用戶在群組中是否有角色

`is_user_in_course_group_or_admin_group()` 組合檢查（成員或管理員）

資料庫統計

項目數量

核心表格30+

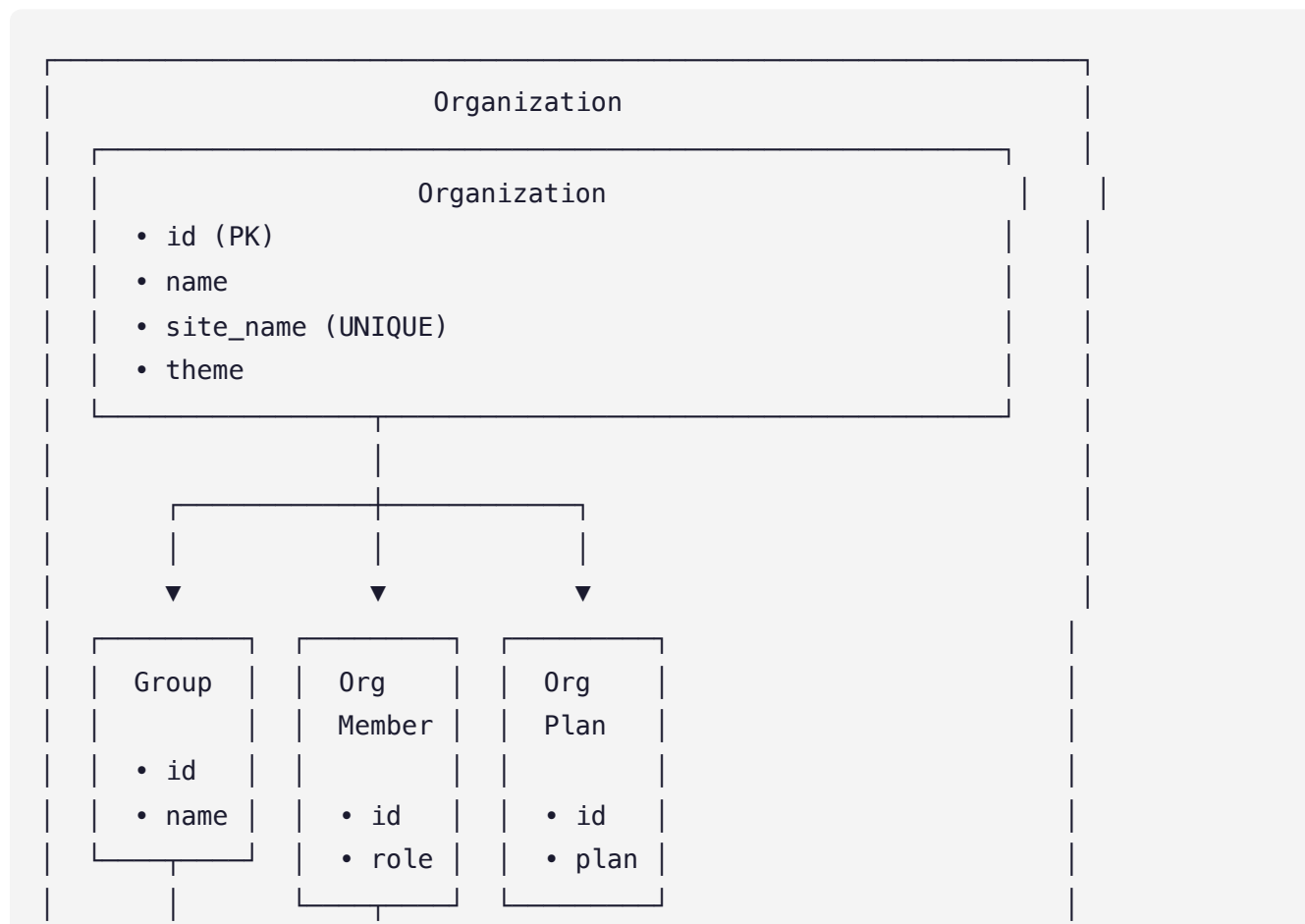
遷移檔案37

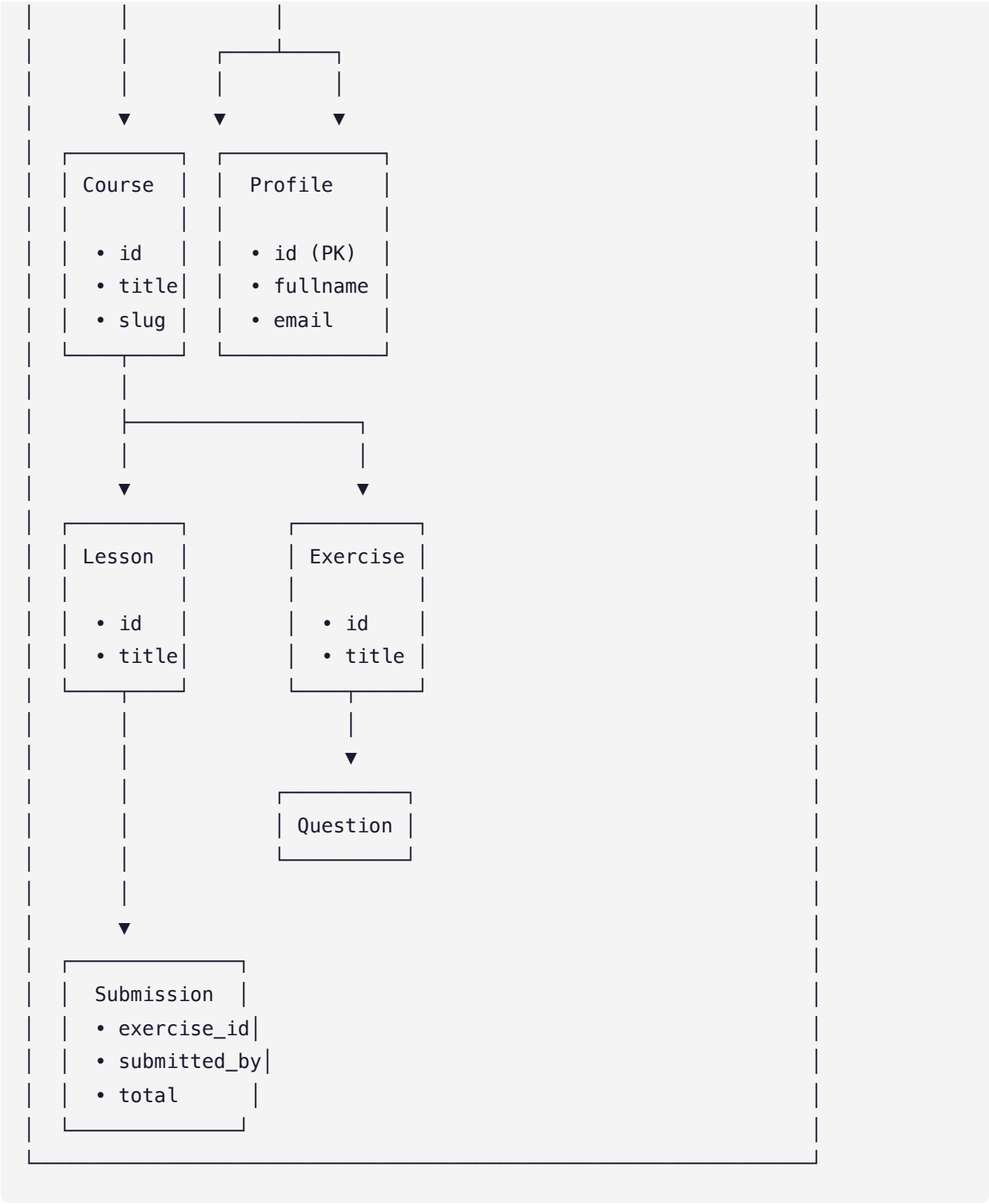
儲存函數10+

RLS 策略50+

資料庫結構

ER 圖（簡化版）





核心表格

1. Profile（用戶資料）

儲存用戶的基本資訊，與 Supabase Auth 的 `auth.users` 表關聯。

```
CREATE TABLE public.profile (
  id uuid REFERENCES auth.users(id) PRIMARY KEY,
  fullname text,
  username text UNIQUE,
  email text,
  avatar_url text,
  created_at timestamp with time zone DEFAULT now(),
  updated_at timestamp with time zone DEFAULT now()
);
```

欄位說明：

欄位類型說明

`id` uuid主鍵，關聯 `auth.users`
`fullname` text用戶全名
`username` text用戶名（唯一）
`email` text電子郵箱
`avatar_url` text頭像 URL
`created_at` timestamp建立時間
`updated_at` timestamp更新時間

2. Organization（組織）

儲存組織資訊，支援多租戶。

```
CREATE TABLE public.organization (
  id uuid DEFAULT uuid_generate_v4() PRIMARY KEY,
  name text NOT NULL,
  site_name text UNIQUE NOT NULL,
  description text,
  logo text DEFAULT '',
  landing_page jsonb DEFAULT '{} '::jsonb,
  theme text DEFAULT 'default',
  created_at timestamp with time zone DEFAULT now(),
  updated_at timestamp with time zone DEFAULT now()
);
```

欄位說明：

欄位類型說明

id uuid主鍵
 name text組織名稱
 site_name text站點名稱（用於子域名，唯一）
 description text組織描述
 logo textLogo URL
 landing_page jsonb落地頁配置
 theme text主題色

3. Organization Member（組織成員）

記錄組織成員及其角色。

```
CREATE TABLE public.organizationmember (
  id uuid DEFAULT uuid_generate_v4() PRIMARY KEY,
  organization_id uuid REFERENCES public.organization(id),
  profile_id uuid REFERENCES public.profile(id),
  role_id bigint REFERENCES public.role(id),
  created_at timestamp with time zone DEFAULT now(),
  updated_at timestamp with time zone DEFAULT now()
);
```

欄位說明：

欄位類型說明

id uuid主鍵
 organization_id uuid所屬組織 ID
 profile_id uuid用戶 ID
 role_id bigint角色 ID (ADMIN/TEACHER/STUDENT)

4. Role（角色）

定義系統角色。

```
CREATE TABLE public.role (
  id bigint GENERATED ALWAYS AS IDENTITY PRIMARY KEY,
  name text NOT NULL
);
```

預設角色：

ID名稱說明

1ADMIN組織管理員

2TEACHER教師

3STUDENT學生

5. Group（群組）

課程所屬的群組，用於權限管理。

```
CREATE TABLE public."group" (
  id uuid DEFAULT uuid_generate_v4() PRIMARY KEY,
  name text,
  organization_id uuid REFERENCES public.organization(id),
  created_at timestamp with time zone DEFAULT now(),
  updated_at timestamp with time zone DEFAULT now()
);
```

6. Group Member（群組成員）

記錄群組成員，用於課程權限控制。

```
CREATE TABLE public.groupmember (
  id uuid DEFAULT uuid_generate_v4() PRIMARY KEY,
  group_id uuid REFERENCES public."group"(id),
  profile_id uuid REFERENCES public.profile(id),
  role_id bigint REFERENCES public.role(id),
  assigned_student_id text,
  created_at timestamp with time zone DEFAULT now(),
```

```
updated_at timestamp with time zone DEFAULT now()
);
```

7. Course (課程)

儲存課程資訊。

```
CREATE TABLE public.course (
  id uuid DEFAULT uuid_generate_v4() PRIMARY KEY,
  title character varying NOT NULL,
  description character varying NOT NULL,
  overview character varying DEFAULT 'Welcome to this amazing course 🚀',
  logo text DEFAULT '' NOT NULL,
  banner_image text,
  slug character varying,
  group_id uuid REFERENCES public."group"(id),
  is_template boolean DEFAULT true,
  is_published boolean DEFAULT false,
  is_certificate_downloadable boolean DEFAULT false,
  certificate_theme text,
  cost bigint DEFAULT 0,
  currency character varying DEFAULT 'NGN' NOT NULL,
  metadata jsonb DEFAULT '{"goals": "", "description": "", "requirements": ""}' NOT NULL,
  status text DEFAULT 'ACTIVE' NOT NULL,
  created_at timestamp with time zone DEFAULT now(),
  updated_at timestamp with time zone DEFAULT now()
);
```

欄位說明：

欄位類型說明

id uuid主鍵

title varchar課程標題

description varchar課程描述

overview varchar課程概述

logo textLogo URL

banner_image text橫幅圖片 URL

slug varcharURL 友善標識

group_id uuid所屬群組 ID

`is_published` boolean 是否已發佈

`cost` bigint 課程費用

`currency` varchar 貨幣

`metadata` jsonb 元資料（目標、需求等）

`status` text 狀態（ACTIVE/ARCHIVED）

8. Lesson（課時）

儲存課程的課時內容。

```
CREATE TABLE public.lesson (  
  id uuid DEFAULT uuid_generate_v4() PRIMARY KEY,  
  title character varying NOT NULL,  
  course_id uuid REFERENCES public.course(id),  
  "order" integer DEFAULT 0,  
  is_unlocked boolean DEFAULT true,  
  created_at timestamp with time zone DEFAULT now(),  
  updated_at timestamp with time zone DEFAULT now()  
);
```

9. Exercise（練習）

儲存課時的練習題組。

```
CREATE TABLE public.exercise (  
  id uuid DEFAULT uuid_generate_v4() PRIMARY KEY,  
  title character varying NOT NULL,  
  lesson_id uuid REFERENCES public.lesson(id),  
  created_at timestamp with time zone DEFAULT now(),  
  updated_at timestamp with time zone DEFAULT now()  
);
```

10. Question（題目）

儲存練習的題目。

```
CREATE TABLE public.question (
  id uuid DEFAULT uuid_generate_v4() PRIMARY KEY,
  exercise_id uuid REFERENCES public.exercise(id),
  title character varying NOT NULL,
  description text,
  type character varying,
  points integer DEFAULT 0,
  "order" integer DEFAULT 0,
  created_at timestamp with time zone DEFAULT now(),
  updated_at timestamp with time zone DEFAULT now()
);
```

11. Submission (提交)

儲存學生的練習提交記錄。

```
CREATE TABLE public.submission (
  id uuid DEFAULT uuid_generate_v4() PRIMARY KEY,
  exercise_id uuid REFERENCES public.exercise(id),
  submitted_by uuid REFERENCES public.groupmember(id),
  status_id bigint REFERENCES public.submissionstatus(id),
  total integer DEFAULT 0,
  feedback text,
  created_at timestamp with time zone DEFAULT now(),
  updated_at timestamp with time zone DEFAULT now()
);
```

12. Submission Status (提交狀態)

定義提交的狀態。

```
CREATE TABLE public.submissionstatus (
  id bigint GENERATED ALWAYS AS IDENTITY PRIMARY KEY,
  name text NOT NULL
);
```

預設狀態：

ID名稱說明

1NOT_SUBMITTED未提交

2SUBMITTED已提交

3IN_PROGRESS評分中

4GRADED已評分

13. Community Question（社群問題）

儲存社群討論區的問題。

```
CREATE TABLE public.community_question (  
  id uuid DEFAULT uuid_generate_v4() PRIMARY KEY,  
  title character varying NOT NULL,  
  body text,  
  course_id uuid REFERENCES public.course(id),  
  org_id uuid REFERENCES public.organization(id),  
  author_id uuid REFERENCES public.profile(id),  
  created_at timestamp with time zone DEFAULT now(),  
  updated_at timestamp with time zone DEFAULT now()  
);
```

14. Community Answer（社群回答）

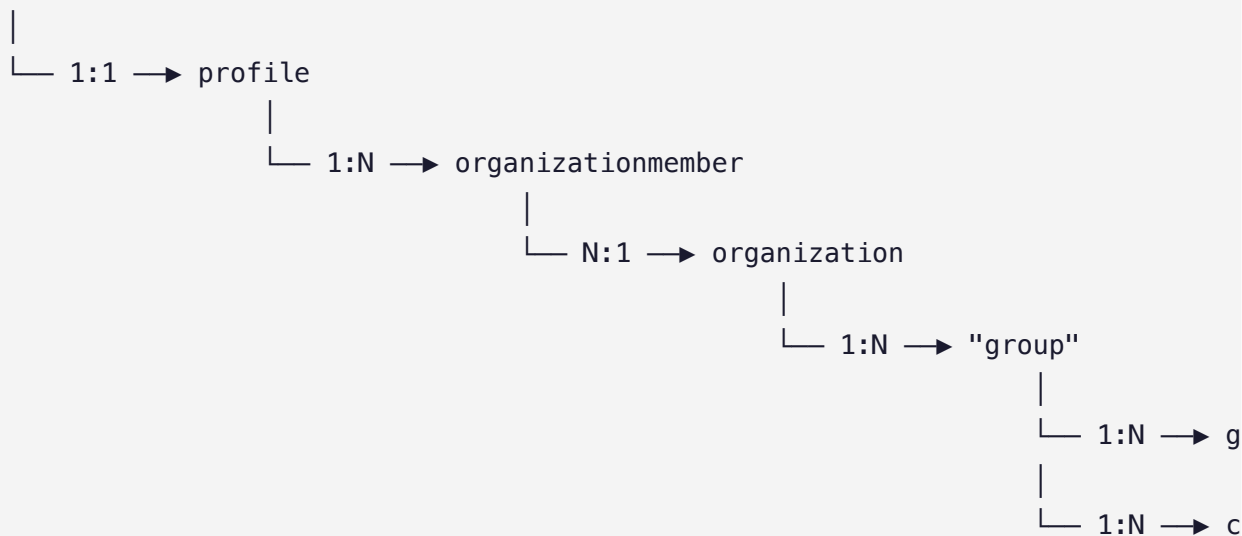
儲存社群問題的回答。

```
CREATE TABLE public.community_answer (  
  id uuid DEFAULT uuid_generate_v4() PRIMARY KEY,  
  body text NOT NULL,  
  question_id uuid REFERENCES public.community_question(id),  
  author_id uuid REFERENCES public.profile(id),  
  created_at timestamp with time zone DEFAULT now(),  
  updated_at timestamp with time zone DEFAULT now()  
);
```

關聯關係

核心關聯

auth.users (Supabase Auth)



外鍵約束

表格欄位參照

profile id auth.users(id)

organizationmember organization_id organization(id)

organizationmember profile_id profile(id)

organizationmember role_id role(id)

group organization_id organization(id)

groupmember group_id group(id)

groupmember profile_id profile(id)

groupmember role_id role(id)

```

course group_id group(id)
lesson course_id course(id)
exercise lesson_id lesson(id)
question exercise_id exercise(id)
submission exercise_id exercise(id)
submission submitted_by groupmember(id)
submission status_id submissionstatus(id)
community_question course_id course(id)
community_question org_id organization(id)
community_question author_id profile(id)
community_answer question_id community_question(id)
community_answer author_id profile(id)

```

RLS 策略

概述

ClassroomIO 使用 Supabase 的 Row Level Security (RLS) 實現數據隔離。所有公開表格都啟用了 RLS。

核心策略示例

Profile 策略

```

-- 用戶可以查看所有公開資料
CREATE POLICY "公開查看用戶資料" ON public.profile
  FOR SELECT USING (true);

-- 用戶只能修改自己的資料
CREATE POLICY "用戶修改個人資料" ON public.profile
  FOR UPDATE USING (id = auth.uid());

-- 用戶只能刪除自己的帳號
CREATE POLICY "用戶刪除帳號" ON public.profile
  FOR DELETE USING (id = auth.uid());

```

Course 策略

-- 用戶只能查看自己所屬組織的課程

```
CREATE POLICY "查看組織課程" ON public.course
FOR SELECT USING (
    group_id IN (
        SELECT group_id FROM public.groupmember
        WHERE profile_id = auth.uid()
    )
);
```

-- 組織管理員可以建立課程

```
CREATE POLICY "管理員建立課程" ON public.course
FOR INSERT WITH CHECK (
    EXISTS (
        SELECT 1 FROM public.organizationmember om
        JOIN public.role r ON r.id = om.role_id
        WHERE om.profile_id = auth.uid()
        AND r.name = 'ADMIN'
    )
);
```

Submission 策略

-- 學生可以查看自己的提交

```
CREATE POLICY "學生查看自己的提交" ON public.submission
FOR SELECT USING (
    submitted_by IN (
        SELECT id FROM public.groupmember
        WHERE profile_id = auth.uid()
    )
);
```

-- 教師可以查看課程的所有提交

```
CREATE POLICY "教師查看課程提交" ON public.submission
FOR SELECT USING (
    exercise_id IN (
        SELECT e.id FROM public.exercise e
        JOIN public.lesson l ON l.id = e.lesson_id
        JOIN public.course c ON c.id = l.course_id
        JOIN public.groupmember gm ON gm.group_id = c.group_id
        WHERE gm.profile_id = auth.uid()
        AND gm.role_id IN (1, 2) -- ADMIN or TEACHER
    )
);
```

```
)
);
```

策略檢查

```
-- 列出所有啟用 RLS 的表格
SELECT tablename FROM pg_tables
WHERE schemaname = 'public'
AND tablename IN (
  SELECT tablename FROM pg_tables
  WHERE schemaname = 'public'
);

-- 查看特定表格的 RLS 策略
SELECT * FROM pg_policies WHERE tablename = 'course';
```

遷移管理

遷移檔案位置

```
supabase/migrations/
├─ 20231115082347_remote_schema.sql      # 初始 schema
├─ 20231118210545_profile.sql            # 用戶資料
├─ 20231207061507_update_get_courses.sql
├─ 20231213104402_course_metadata.sql
├─ ...
├─ 20251205045311_modify_profile_policy.sql
└─ (共 37 個遷移檔案)
```

執行遷移

本地執行所有遷移

```
supabase db reset
```

執行特定遷移

```
supabase migration up
```

連結到雲端項目

```
supabase link --project-ref
```

推送遷移到雲端

```
supabase db push --include-all
```

建立新遷移

```
supabase migration new
```

遷移最佳實踐

1. **備份先行:** 執行遷移前先備份資料庫
2. **測試環境:** 先在本地或測試環境執行

3. 向前兼容: 避免破壞性變更 (如刪除欄位)
4. 版本控制: 所有遷移檔案納入版本控制

常用查詢

取得用戶的所有課程

```
SELECT c.*, gm.role_id
FROM public.course c
JOIN public."group" g ON g.id = c.group_id
JOIN public.groupmember gm ON gm.group_id = g.id
WHERE gm.profile_id = ''
AND c.status = 'ACTIVE'
ORDER BY c.created_at DESC;
```

取得課程的課時與進度

```
SELECT
  l.id,
  l.title,
  l."order",
  CASE WHEN lc.is_complete THEN true ELSE false END as is_completed
FROM public.lesson l
LEFT JOIN public.lesson_completion lc
  ON lc.lesson_id = l.id AND lc.profile_id = ''
WHERE l.course_id = ''
ORDER BY l."order";
```

取得練習的提交統計

```
SELECT
  e.id as exercise_id,
  e.title as exercise_title,
  COUNT(s.id) as total_submissions,
  COUNT(CASE WHEN ss.name = 'GRADED' THEN 1 END) as graded_count,
  AVG(CASE WHEN ss.name = 'GRADED' THEN s.total END) as avg_score
```

```

FROM public.exercise e
LEFT JOIN public.submission s ON s.exercise_id = e.id
LEFT JOIN public.submissionstatus ss ON ss.id = s.status_id
WHERE e.lesson_id = ''
GROUP BY e.id, e.title;

```

取得組織的社群問題

```

SELECT
  cq.*,
  p.fullname as author_name,
  p.avatar_url as author_avatar,
  c.title as course_title,
  (SELECT COUNT(*) FROM public.community_answer ca WHERE ca.question_id = cq.id)
FROM public.community_question cq
JOIN public.profile p ON p.id = cq.author_id
LEFT JOIN public.course c ON c.id = cq.course_id
WHERE cq.org_id = ''
ORDER BY cq.created_at DESC;

```

取得學生的學習分析

```

SELECT
  p.fullname,
  p.email,
  COUNT(DISTINCT c.id) as enrolled_courses,
  COUNT(DISTINCT CASE WHEN lc.is_complete THEN l.id END) as completed_lessons,
  COUNT(DISTINCT s.id) as total_submissions,
  AVG(CASE WHEN ss.name = 'GRADED' THEN s.total END) as avg_score
FROM public.profile p
JOIN public.groupmember gm ON gm.profile_id = p.id
JOIN public."group" g ON g.id = gm.group_id
JOIN public.course c ON c.group_id = g.id
LEFT JOIN public.lesson l ON l.course_id = c.id
LEFT JOIN public.lesson_completion lc ON lc.lesson_id = l.id AND lc.profile_id = p.id
LEFT JOIN public.groupmember gm2 ON gm2.profile_id = p.id
LEFT JOIN public.submission s ON s.submitted_by = gm2.id
LEFT JOIN public.submissionstatus ss ON ss.id = s.status_id

```

```
WHERE p.id = ''
GROUP BY p.id, p.fullname, p.email;
```

附錄

索引建議

為提升查詢效能，建議為以下欄位建立索引：

```
-- 課程查詢
CREATE INDEX idx_course_group_id ON public.course(group_id);
CREATE INDEX idx_course_status ON public.course(status);
CREATE INDEX idx_course_slug ON public.course(slug);

-- 課時查詢
CREATE INDEX idx_lesson_course_id ON public.lesson(course_id);

-- 練習查詢
CREATE INDEX idx_exercise_lesson_id ON public.exercise(lesson_id);

-- 題目查詢
CREATE INDEX idx_question_exercise_id ON public.question(exercise_id);

-- 提交查詢
CREATE INDEX idx_submission_exercise_id ON public.submission(exercise_id);
CREATE INDEX idx_submission_submitted_by ON public.submission(submitted_by);

-- 群組成員查詢
CREATE INDEX idx_groupmember_group_id ON public.groupmember(group_id);
CREATE INDEX idx_groupmember_profile_id ON public.groupmember(profile_id);

-- 社群查詢
CREATE INDEX idx_community_question_org_id ON public.community_question(org_id);
CREATE INDEX idx_community_answer_question_id ON public.community_answer(question
```

備份與恢復

完整備份

```
supabase db dump --db-url postgresql://postgres:@:5432/postgres > backup.sql
```

只備份 Schema

```
supabase db dump --schema-only > schema.sql
```

只備份資料

```
supabase db dump --data-only > data.sql
```

恢復

```
psql postgresql://postgres:@:5432/postgres < backup.sql
```